NAME

delta − differences between two files

SYNOPSIS

delta [*options*] *file1 file2*

DESCRIPTION

delta computes the Longest Common Subsequence (LCS) of *file1* and *file2*, and prints their differences as an edit script. With **-l** it prints only the LCS length.

delta is a simplified companion of **compare**(1): it always reports differences (there is no correspondence mode), it always uses the Hunt-Szymanski algorithm, and it compares records rather than code points.

Each difference is introduced by a header line:

<<<<< n *file1* [*begin*:*end*] <<<<<
(records present only in *file1*)
>>>>> n *file2* [*begin*:*end*] >>>>>
(records present only in *file2*)

where *n* is the one-based difference number and *[begin:end)* the half-open range of records involved. A final line reports the LCS length, followed (when compiled with duration reporting enabled) by a timing line such as *Hunt-Szymanski: 28 operations in 9 µs @3.111 MHz*.

Each input line is one record; with **-w** each whitespace-separated word is one record.

OPTIONS

-b Ignore whitespace: collapse each run of whitespace to a single space before comparing, and expand the reported spans back to the original positions.

-E Print the host environment (locale, processor, and memory) and exit.

-h Print a usage synopsis and exit.

-i Perform a case-insensitive comparison.

-j *join* Coalesce nearby differences that lie within *join* records of each other.

-l Report only the LCS length; skip reconstructing the differences.

-p *prefix*
Print *prefix* lines of context before each difference.

-s *suffix*
Print *suffix* lines of context after each difference.

-t *tabsize*
Set the tab size used to expand tab characters in the parallel **-y** p output (default 8).

-v Enable verbose output: print the LCS length, the start and finish wall-clock times, and a timing summary; by default only the differences are shown.

-V Print the version and exit.

-w Split each line into whitespace-separated words and compare word by word.

-W *width*
Set the per-side content width used by the parallel **-y** p format (default 80, minimum 5). Content is truncated to this width, so the total output line is 1 + 2 x width characters.

-y *s|p* Select the output format: **s** (sequential, the default: one side after the other) or **p** (parallel: side-by-side columns).

The **-h**, **-V**, and **-E** switches are early-exit switches: each prints its report and terminates without comparing files.

EXIT STATUS

0 Success (including **-h**, **-V**, and **-E**).

1 Error: a command-line parse failure, a missing file, or an algorithm failure.

ENVIRONMENT

The locale (in particular **LANG**, **LC_ALL**, and **LC_NUMERIC**) controls the thousands separator used in the reported counts and the locale name shown by **-E**. On a system whose locale defines digit grouping, counts are printed with comma separators (for example **LCS Length = 4,591,960**); otherwise they are printed ungrouped.

EXAMPLES

Compare two files line by line and show the differences:

```
delta old.txt new.txt
```

Compare two files word by word, ignoring case:

```
delta -i -w a.txt b.txt
```

Report only the LCS length of two files:

```
delta -l a.txt b.txt
```

SEE ALSO

compare(1), **diff**(1), **cmp**(1)